

Unex App Summary

One-page repo-based overview for the app in `/Users/ke/lima-workspace/unions`

What It Is

Unex is an open-source ops platform for Unison programs. The repo describes and implements a BEAM-based system that provides durable storage, encrypted secrets, clustering, and content-addressed code execution through an authenticated HTTP API plus a matching Unison ability library.

Who It's For

Primary persona: developers or platform operators who need to run, deploy, and operate Unison applications while keeping storage, secrets, and service wiring outside the app code.

What It Does

- Runs Unison source directly or compiles it to cached `.uc` bytecode before execution.
- Exposes authenticated HTTP endpoints for databases, tables, cells, transactions, config, blobs, scratch, logs, services, and bytecode.
- Persists structured storage in Mnesia and stores binary blobs on the filesystem under the configured data directory.
- Encrypts config secrets with AES-256-GCM before storing them in Mnesia.
- Caches bytecode by SHA256 in ETS and resolves missing hashes from connected cluster peers.
- Supports named service release and invocation by mapping service names to bytecode hashes.

How It Works

Runtime: `Unex.Application` starts `HashCache`, `SyncServer`, `Services.Registry`, `Scratch`, and `Log`, and conditionally starts `PeerConnector` and the Bandit API server.

API layer: Plug Router parses JSON, enforces bearer auth on all routes except `/health`, and dispatches to controllers for storage, services, config, blobs, scratch, log, and bytecode.

Execution flow: `Unex.eval/2` and `Unex.compile_and_run/2` create an isolated workspace, write source into it, and use UCM to run source or compile then run bytecode.

Data flow: Unison programs use the library in `unison/`; handlers make authenticated HTTP calls to the Elixir API; data lands in Mnesia or filesystem blobs; compiled bytecode is cached in ETS and can be fetched from peers for remote execution.

How To Run

- Install prerequisites: Elixir 1.17+ and the UCM binary on `PATH`.
- From the repo root, run `mix deps.get`.
- Start the app with `mix unex.start`.
- Use the printed secret, or set `UNEX_SECRET` to persist it across restarts.
- Verify the API with `curl -s localhost:4040/health`.
- For clustered startup, the repo documents `UNEX_NODE`, `UNEX_COOKIE`, `UNEX_PORT`, and `UNEX_PEERS`.

Repo Gaps

Pricing, deployment packaging beyond a Mix release, and a named commercial user segment were Not found in repo.